Chapter 2 — Atomic Structure

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Fundamental particles and atomic identity

Q.1 Explain atomic number, mass number and nucleon number, and describe the behaviour of fundamental particles in an electric field.

An atom contains protons, neutrons and electrons. The atomic number, Z, is the number of protons and identifies the element. The mass number, A, is the total number of protons and neutrons, also called the nucleon number. In a neutral atom, electrons equal protons; in an ion, electrons are lost or gained.

In an electric field, electrons are deflected towards the positive plate because they are negatively charged. Protons are deflected towards the negative plate but much less because they are much heavier. Neutrons are undeflected because they carry no charge.

Particle	Relative charge	Relative mass	Behaviour in electric field
Electron	–1	1/1836	Deflected strongly towards positive plate
Proton	+1	1	Deflected slightly towards negative plate
Neutron	0	1	No deflection

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Important Conceptual Short Questions — Topic 1

1. Why does atomic number identify an element more reliably than mass number?

All atoms of an element have the same number of protons, but they may have different numbers of neutrons. Therefore atomic number is fixed for an element, while mass number can vary among isotopes.

2. How can the number of neutrons in an ion be calculated from A and Z?

The number of neutrons is A – Z. Ionic charge changes only the number of electrons, so it does not affect the neutron count.

3. Why are electrons deflected more strongly than protons in the same electric field?

Electrons have the same magnitude of charge as protons but a much smaller mass. The same electric force produces a much greater acceleration and deflection for electrons.

Topic 2 — Atomic spectra and energy levels

Q.2 Explain atomic spectra and how they support the idea of quantized energy levels.

When atoms absorb energy, electrons move to higher energy levels. When they return to lower levels, energy is emitted as light of definite wavelengths. This produces a line spectrum rather than a continuous spectrum. Each element has a unique line spectrum, like a fingerprint.

The hydrogen spectrum shows that electrons can exist only in certain allowed energy levels. The Lyman series lies in the ultraviolet region, the Balmer series in the visible region, and the Paschen series in the infrared region. The lines arise from transitions between fixed energy levels.

Important Conceptual Short Questions — Topic 2

1. Why do atoms produce line spectra instead of continuous spectra?

Electrons in atoms can occupy only certain allowed energy levels. When they jump between these levels, they emit or absorb photons of fixed energy, so only definite wavelengths appear.

2. How can atomic spectra be used to identify an element?

Each element has a unique set of energy levels and therefore a unique pattern of spectral lines. Matching an unknown spectrum with a known pattern identifies the element.

3. Why does the Balmer series appear in the visible region?

The Balmer series is produced when electrons fall from higher levels to n = 2. The energy differences for these transitions correspond to photons in the visible region.

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Topic 3 — Successive ionization energies and shells

Q.3 How do successive ionization energies provide evidence for electronic shells?

Successive ionization energies are the energies required to remove electrons one by one from a gaseous atom or ion. They generally increase because each electron is removed from an increasingly positive species. A very large jump occurs when an electron is removed from a new inner shell.

For example, sodium shows a large jump after the first electron is removed. This shows that sodium has one electron in its outer shell and a stable inner core. The pattern of jumps gives evidence for the number of electrons in each shell.

Important Conceptual Short Questions — Topic 3

1. Why does a large jump in successive ionization energies indicate a new shell?

Electrons in an inner shell are closer to the nucleus and less shielded. Removing one requires much more energy, so a sudden jump shows that the outer shell has been emptied.

2. How can successive ionization energies help identify the group of an element?

The number of electrons removed before the first large jump equals the number of outer-shell electrons. For main-group elements, this indicates the group number.

3. Why does each successive ionization energy increase even within the same shell?

After each electron is removed, the remaining electrons experience a greater effective nuclear charge and less repulsion. They are held more strongly, so more energy is needed for the next removal.

Topic 4 — Quantum numbers

Q.4 Describe the four quantum numbers and the information they give about an electron.

The principal quantum number, n, gives the main shell, size and energy. The azimuthal quantum number, l, gives the subshell shape: s, p, d or f. The magnetic quantum number, m, gives the orientation of an orbital in space. The spin quantum number, s, gives the spin of the electron, +½ or –½.

For a given n, l can have values from 0 to n–1. For a given l, m can have values from –l to +l. Therefore an s subshell has 1 orbital, p has 3, d has 5 and f has 7. No two electrons in an atom can have the same set of four quantum numbers.

Quantum number	Symbol	Information
Principal	n	Shell, size and energy
Azimuthal	l	Subshell shape
Magnetic	m	Orbital orientation
Spin	s	Electron spin direction

Important Conceptual Short Questions — Topic 4

1. Why can a p subshell contain three orbitals?

For a p subshell, l = 1, so m can be –1, 0 or +1. These three values correspond to three orientations, usually called p_x, p_y and p_z.

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2. How does the azimuthal quantum number determine subshell shape?

The value of l defines the angular distribution of the electron cloud. l = 0 gives a spherical s orbital, l = 1 gives dumb-bell p orbitals, and higher values give d and f shapes.

3. Why can an orbital hold only two electrons?

An orbital is defined by n, l and m. The only remaining quantum number is spin, which has two values, +½ and –½. By Pauli's exclusion principle, only two electrons with opposite spins can occupy it.

Topic 5 — Shapes of orbitals and electronic configuration

Q.5 Describe the shapes of s, p and d orbitals and explain the rules used to write electronic configurations.

An s orbital is spherical and non-directional. A p orbital is dumb-bell shaped and has two lobes with a nodal plane at the nucleus. d orbitals have more complex shapes, mostly four-lobed, and there are five orientations in a d subshell.

Electrons fill orbitals according to three rules. The Aufbau principle states that lower-energy orbitals fill first. Pauli's exclusion principle states that an orbital can hold at most two electrons with opposite spins. Hund's rule states that degenerate orbitals are occupied singly before pairing. Chromium and copper show exceptions because half-filled and fully filled d subshells are especially stable.

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Important Conceptual Short Questions — Topic 5

1. Why do electrons occupy degenerate orbitals singly before pairing?

Singly occupying orbitals with parallel spins reduces electron-electron repulsion and gives a lower-energy arrangement. This is Hund's rule.

2. How does the Aufbau principle explain the filling order 4s before 3d?

In neutral atoms, the 4s orbital has slightly lower energy than 3d at the point of filling, so electrons enter 4s first. After filling, 3d becomes lower in energy, which is why transition-metal ions lose 4s electrons first.

3. Why is the electronic configuration of chromium exceptional?

Chromium is [Ar] $3d^5$ $4s^1$ rather than [Ar] $3d^4$ $4s^2$. A half-filled d subshell gives extra stability, so one 4s electron is promoted to 3d.

Topic 6 — Ions, free radicals and semiconductors

Q.6 Explain the formation of ions and free radicals, and describe p-type and n-type semiconductors.

Ions form when atoms lose or gain electrons. Cations are positive and anions are negative. Free radicals are species with an unpaired electron; they are usually very reactive. Electronic configuration explains why some species form stable ions while others remain radicals.

Semiconductors such as silicon have conductivity between conductors and insulators. In n-type semiconductors, silicon is doped with a Group 15 element such as phosphorus, providing extra electrons. In p-type semiconductors, silicon is doped with a Group 13 element such as boron, creating electron holes. These materials are essential in diodes, transistors and solar cells.

Important Conceptual Short Questions — Topic 6

1. How do cations and anions differ in size from their parent atoms?

Cations are smaller because electron loss reduces repulsion and may remove a shell. Anions are larger because electron gain increases repulsion in the valence shell.

2. Why are free radicals usually very reactive?

A free radical has an unpaired electron. It tends to gain, lose or share an electron quickly to achieve a more stable paired arrangement.

3. How does doping increase the conductivity of silicon?

Doping introduces extra charge carriers. Group 15 dopants provide extra electrons for n-type conduction, while Group 13 dopants create holes for p-type conduction.

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